

Modern Embedded Systems Programming

Get beyond the RTOS with actor-based concurrency, hierarchical state machines, graphical modeling, and automatic code generation.

Welcome to the 21st century!

QP Actor Frameworks

Light, open-source frameworks for building responsive and modular embedded code as systems of cooperating **active objects** (actors).



QM Modeling Tool

Free graphical modeling tool for designing and auto-generating embedded code based on hierarchical state machines (UML statecharts) and the QP frameworks.



Quantum Leap
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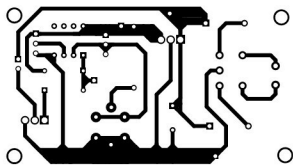


Fig. 3: Actual-size PCB pattern of the LPG detector

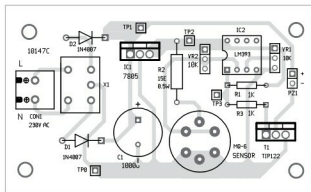


Fig. 4: Component layout of the PCB

the circuit is dual comparator IC LM393 (IC2). It is used to compare two different voltages, namely, reference voltage and MQ-6 gas sensor output voltage.

Reference voltage at non-inverting pin 3 of IC2 is set using potentiometer VR1 to adjust voltage levels based on requirement. LPG sensor (MQ-6) output voltage is fed to inverting pin 2 of IC2.

If reference voltage (pin 3 of IC2) is less than sensor voltage (pin 2 of IC2), output goes low, which means there is no LPG leakage. With low output, T1 remains cut-off and there is no current flow through the buzzer; it does not sound and remains in silence mode.

If reference voltage is greater than sensor voltage, output goes high, which means there is LPG leakage. The high output switches on transistor T1 and the buzzer rings loudly to alert the

people around.

It is very easy to find gas leakages with this circuit, which uses low-cost components and an interactive way to adjust different sensitivity levels, based on customer needs, with the help of potentiometer VR1.

Construction and testing

An actual-size PCB pattern of the LPG detector is shown in Fig. 3 and its component layout in Fig. 4.

After assembling the circuit on a PCB, enclose it in a box with an opening for the

gas to enter. Place the unit near the LPG cylinder or gas stove, within a distance of one metre. Vary preset VR1 to adjust sensitivity of the sensor.

Verify the voltages as per test points table before using the circuit. Now, spray the gas from the bottle (as shown on the left side of author's prototype) towards MQ-6 gas sensor and measure voltage at TP3; it should be high.

If you do not have a gas-filled bottle, place the unit near the gas stove burner and turn it on for a few seconds without igniting. Then, turn the burner off and adjust VR1 until buzzer sounds. **EFY**



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